



Yu Tang

Ph.D. Student

📅 Currently enrolled

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Skills

Programming languages: Python, C++, MATLAB

Deep learning frameworks: PyTorch, TensorFlow, ONNX

Computer vision: OpenCV, object detection, image segmentation, feature extraction and matching, 3D reconstruction

AI training and deployment: familiar with mainstream architectures such as CNNs and Transformers; understands data annotation, data augmentation, model training and tuning, and quantized deployment workflows

Toolchain: Git, Linux, Docker, VSCode, Jupyter; familiar with Conda / venv environment management

Projects

Test Tube Recognition System Based on Global Prediction and Local Cropping Lead researcher 2024/04 - 2025/03

<https://ieeexplore.ieee.org/document/11067950>

An automatic detection and recognition method for robotic test tube manipulation that recognizes newly appearing test tube types in a rack without manually annotated training data.

Proposed a joint "global prediction + local cropping" framework: global prediction estimates the occupancy of each rack slot, and local cropping extracts the test tube image from the corresponding slot region for subsequent processing.

Designed an automated segmentation, clustering, and labeling pipeline that removes the need, inherent in conventional methods, for custom datasets and explicit training for each new test tube type, greatly improving flexibility and deployment efficiency.

Built the experimental platform with a RealSense D405 depth camera and a UFactory xArm Lite6 robot arm, and completed system integration and validation in real-world settings.

Experiments show that the method accurately recognizes previously unseen test tube types in real environments, validating its effectiveness and generalization ability.

Education

The University of Osaka

Research focus: Robot vision

Systems Innovation · Ph.D.

2025/04 - Present

The University of Osaka

Research focus: Robot vision

Systems Innovation · Master's

2022/10 - 2025/03

Beijing Institute of Technology

Optoelectronic Information Science and Technology · Bachelor's

2017/09 - 2022/06

Core courses: Optoelectronic Information Science and Engineering, Digital Image Processing, Signals and Systems, Optical System Design and Fabrication

Extracurricular activities: Vice Head of the Clubs Department, Student Clubs Union; Class Mental Health Representative; Member of the college basketball team

Beijing Jingshan School

Junior high · Senior high

2010/07 - 2017/06